

Continuous functions – 3

Level sets, discontinuities and the fine structure of the graph

Homework

1 сентября 2026 года

Theoretical minimum. Level sets of a continuous function and parity obstructions; one-sided limits and the countability of the set of discontinuities; shadow points; the intermediate value property without continuity (Darboux functions); fixed points on closed subsets of the line; functional relations that force continuity.

1. (IMC 2011) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous. A point x is a *shadow point* if there is $y > x$ with $f(y) > f(x)$. Let $a < b$ be such that all points of (a, b) are shadow points while a and b are not. Prove that a) $f(x) \leq f(b)$ for all $a < x < b$; b) $f(a) = f(b)$.
2. (IMC 2006) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that for all $a < b$ the image $f([a, b])$ is a closed interval of length $b - a$.
3. Let C be a non-empty closed bounded subset of the line and let $f: C \rightarrow C$ be non-decreasing and continuous. Show that f has a fixed point.
4. (RMO) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ have the intermediate value property and be injective on $\mathbb{R} \setminus \mathbb{Q}$. Prove that f is continuous on $\mathbb{R}$.
5. (RMO 2021) Let $f: [a, b] \rightarrow \mathbb{R}$ have the intermediate value property with $f(a)f(b) < 0$. Show that there are α, β with $a < \alpha < \beta < b$ and $f(\alpha) + f(\beta) = f(\alpha)f(\beta)$.
6. (RMO 2019) Let $f: [0, \infty) \rightarrow \mathbb{R}$ be continuous and constant on the non-negative integers. Suppose that whenever $0 \leq a < b < c \leq d$ satisfy $f(a) = f(c)$ and $f(b) = f(d)$, we also have $f(\frac{a+b}{2}) = f(\frac{c+d}{2})$. Prove that f is constant.

7. (RMO 2017) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be surjective and such that convergence of $(f(x_n))$ implies convergence of (x_n) . Prove that f is continuous.
8. (RMO 2022) Let $f: [0, 1] \rightarrow (0, 1)$ be surjective.
- a) Prove that f has at least one point of discontinuity.
 - b) Given that f has a limit at every point of $[0, 1]$, show that it has at least two points of discontinuity.
9. (Putnam 2002) Let $a, b \in (0, \frac{1}{2})$ and let f be a continuous real-valued function with $f(f(x)) = af(x) + bx$ for all $x \in \mathbb{R}$. Prove that $f(0) = 0$.
10. Let $f: [0, 1] \rightarrow \mathbb{R}$ be continuous. Prove that the series $\sum_{n \geq 1} \frac{f(x^n)}{2^n}$ converges for every $x \in [0, 1]$, and find a function satisfying $f(x) = \sum_{n \geq 1} \frac{f(x^n)}{2^n}$ for all $x \in [0, 1]$.